



BUSINESS DATA MANAGEMENT

**Data-Driven Solutions for Financial and
Operational Challenges to Improve Business
Performance at ABC Tile Studio**

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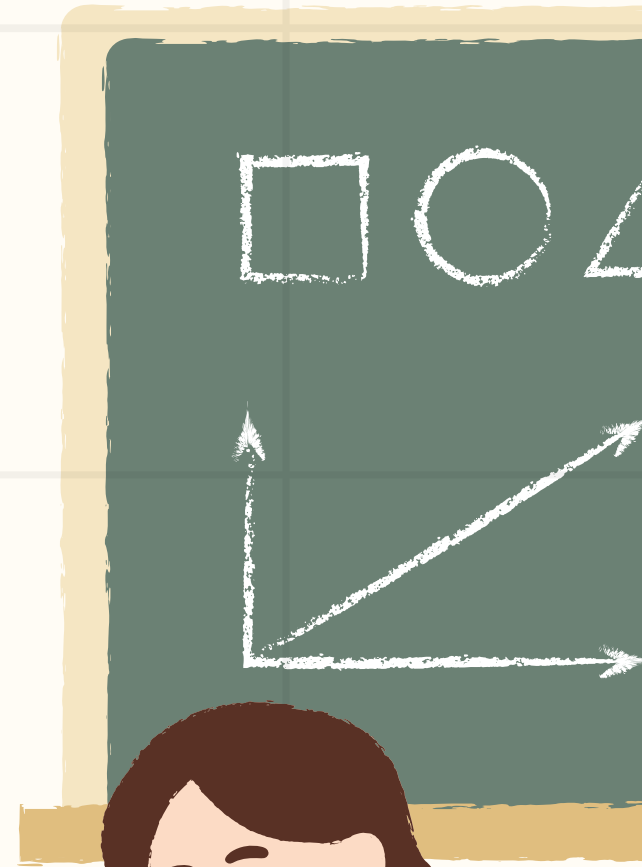


TABLE OF CONTENTS

- 1 Background of Organisation
- 2 Understanding the Problems
- 3 Data Collected and Analysis Methods
- 4 Key Findings and Results
- 5 Proposed Solutions and Expected Outcomes



BACKGROUND OF ORGANISATION

- With over eight years of experience in the industry, ABC Tile Studio has grown into a well-recognized name, trusted by more than 2000 customers.
- Type of business : B2B and B2C
- The company's curated product range includes a diverse selection of tiles, sanitary ware, bath fittings, windows, doors, quartz, slabs, and other building essentials.





UNDERSTANDING THE PROBLEMS

Inefficient Tracking of Client Payments

Problem: Difficulty in monitoring client payments leads to delayed cash inflows, disrupting cash flow and vendor payments.

Aim: Implement a structured payment plan to ensure timely collections and enhance cash flow stability.

Lack of Inventory Insight on Product Movement

Problem: Limited understanding of sales patterns causes inefficiencies, such as overstocking slow-moving tiles and stockouts of fast-moving ones.

Aim: Utilize sales data analysis to optimize inventory decisions and improve STR.

Customer Feedback Utilization

Problem: Insufficient analysis of customer feedback results in missed opportunities to improve service quality and customer satisfaction.

Aim: Analyze feedback data to identify areas of improvement and enhance customer loyalty through targeted actions.



DATA COLLECTED & ANALYSIS METHODS

Client Payment Data

Ledger Reports of 10 clients includes records of debits, credits, and transaction dates, enabling analysis of client payment behaviors and outstanding balances.

Inventory Data

Monthly inventory data for seven types of tiles, detailing inward and outward quantities and values, allowing for an assessment of inventory movement trends and product demand.

Customer feedback

It includes evaluations of service quality, delivery time, staff behavior, store experience, variety of products, product pricing and overall customer satisfaction. This data allows for an analysis of customer perceptions and identification of areas for improvement.

Client Payment Analysis

Monthly credit and debit trends were visualized using bar and line charts to highlight irregularities in payment schedules.

Clients with high outstanding balances were identified to address cash flow challenges effectively.

Inventory Turnover Analysis

Heatmaps and turnover rates were used to pinpoint high-demand and underperforming products.

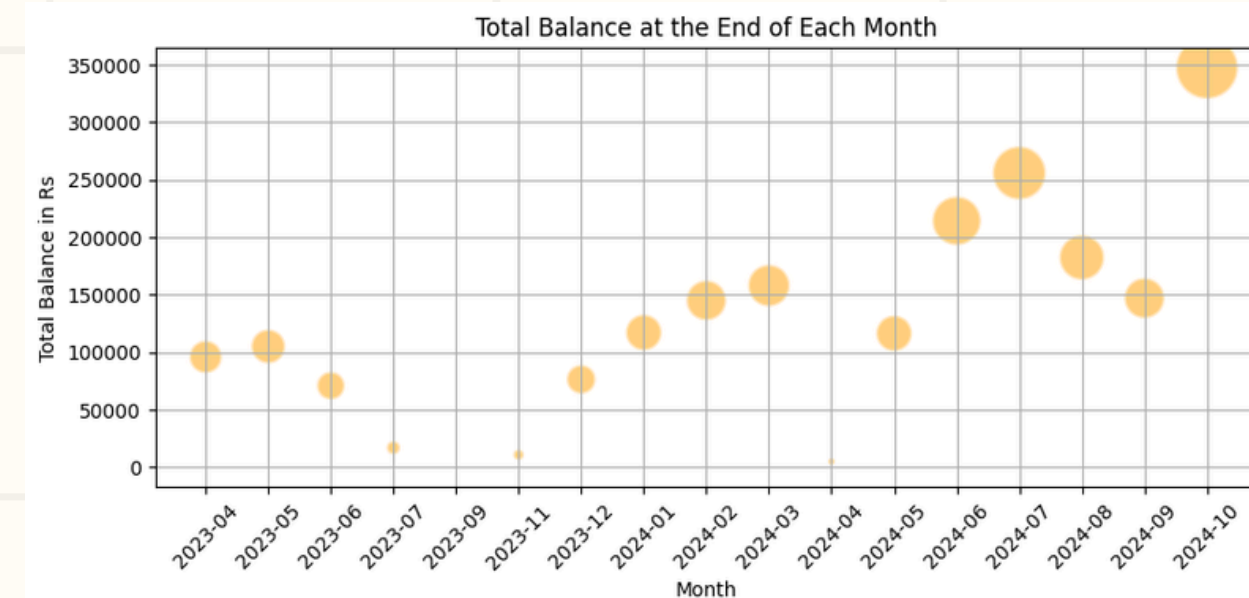
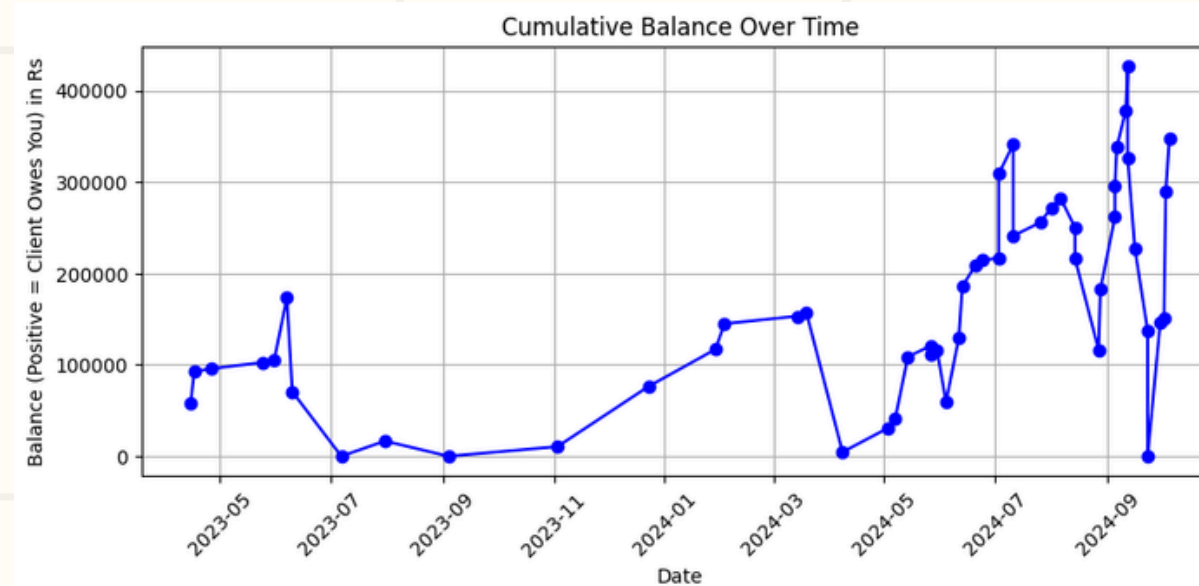
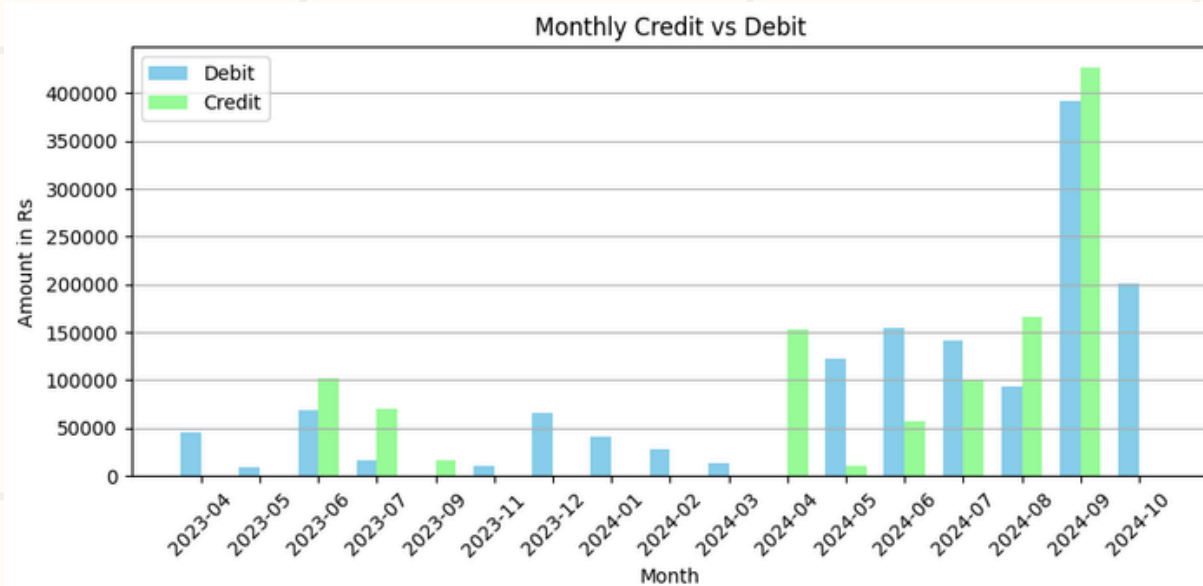
Bubble charts and line graphs helped visualize stock movement and ensure optimal inventory levels.

Customer Feedback Analysis

Pie charts, bar graphs, and box plots summarized customer sentiments across various aspects.

The analysis provided actionable insights to enhance service quality and customer satisfaction.

KEY FINDINGS & RESULTS



Ledger reports of 10 clients were analysed the above graphs are for client- 1

- Bar Chart : Throughout the observed period, credit and debit values fluctuate, with large differences between credit and debit in specific months. This points to a pattern of irregular payment schedules and the inconsistent monthly transactions likely contributes to the challenges in tracking client payments.
- Line Chart : The cumulative balance fluctuates, reflecting intermittent payments and unsettled dues. Despite some reductions in balance after high spikes, the trend remains volatile, emphasizing the need for a more structured payment system. The graph also shows periods of partial payments, where the balance decreases but does not reach zero, indicating a persistent debt.
- Bubble Chart : The trend suggests that balances are either being paid inconsistently or are carried over month to month without resolution. The size of the bubbles represents the magnitude of the closing balance at the end of each month.

KEY FINDINGS & RESULTS

- High Volatility in Demand:

Tiles like "600x1200" and "24x12" exhibit frequent spikes and drops in inventory levels, reflecting fluctuating demand and restocking challenges.

- Stockouts for Popular Items:

Persistent stockouts were observed for tiles like "800x1600," highlighting unmet demand and lost sales opportunities.

- Overstocking of Niche Items:

Tiles such as "800x3000" show consistent but low demand, resulting in excess inventory and higher holding costs.

- Stable Demand for Moderate Items:

Tiles like "100x300" and "1200x1800" have steady but moderate demand, with occasional dips indicating gradual growth.

- Need for Seasonal Planning:

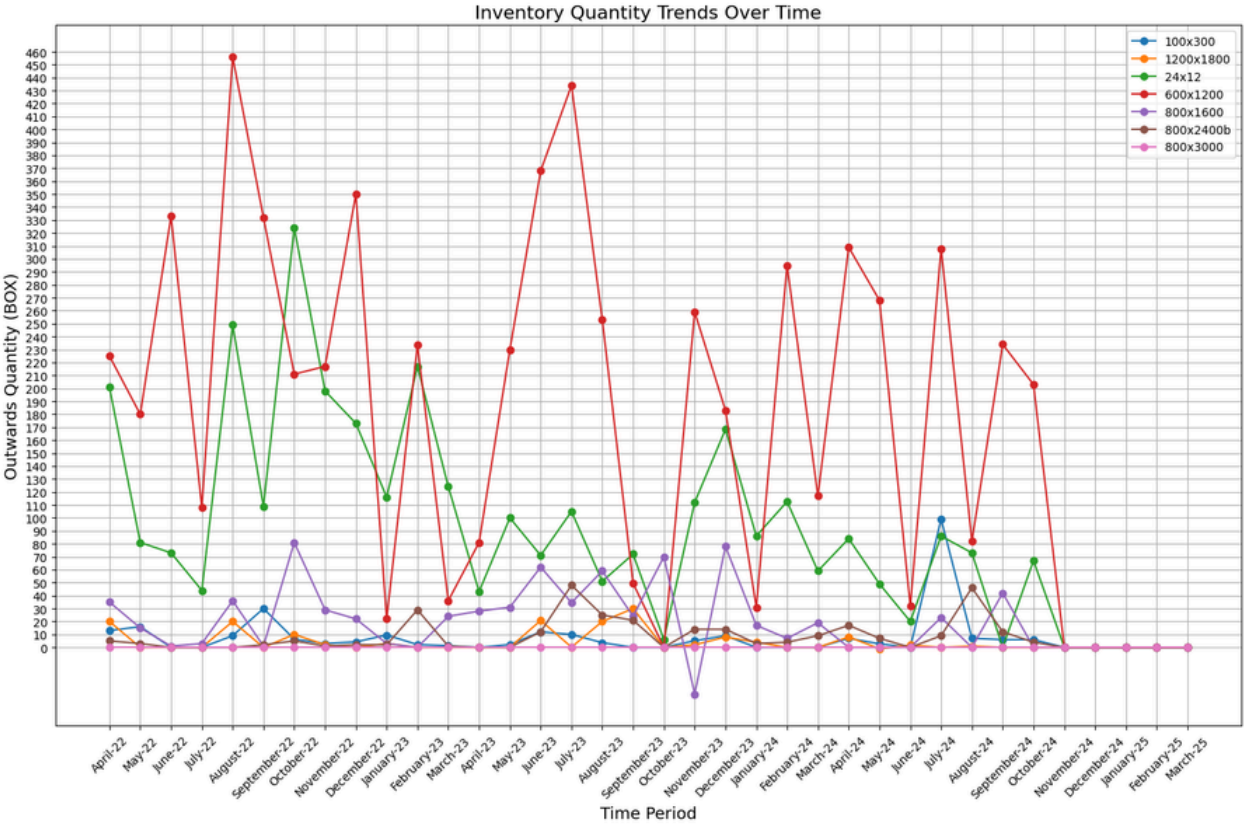
Irregular peaks for "600x1200" suggest seasonal or project-based demand fluctuations that require better forecasting.

- Turnover Insights:

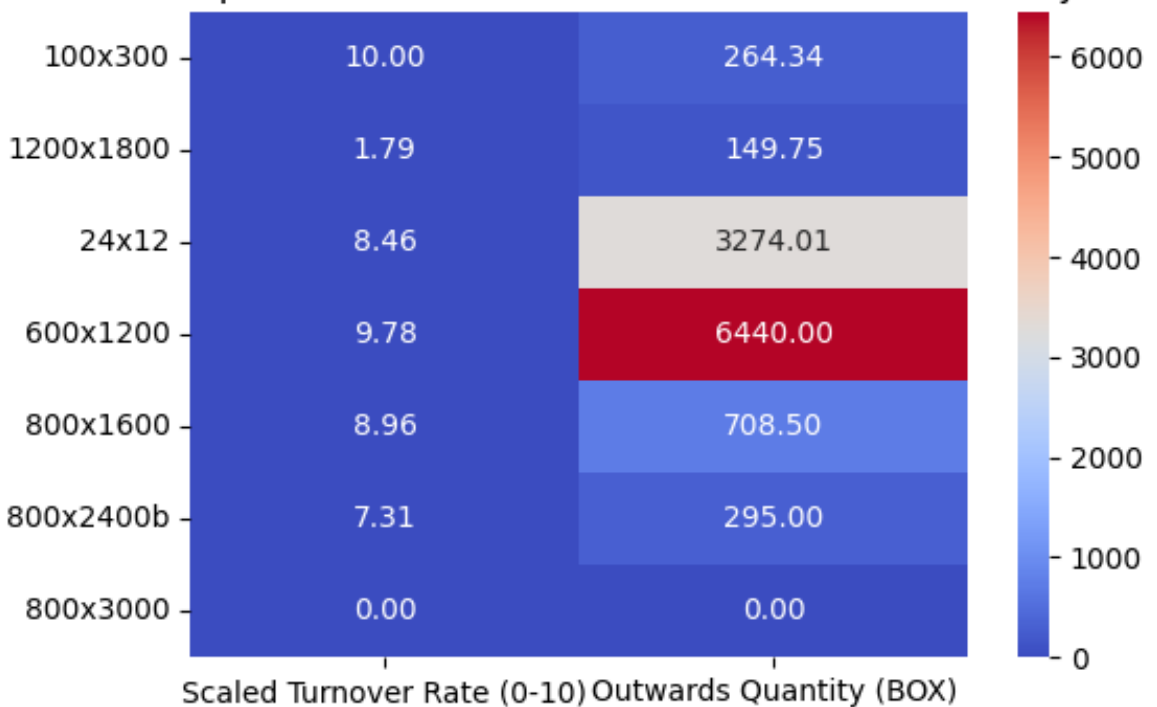
High turnover rates for tiles like "100x300," "600x1200," and "24x12" confirm their popularity, while lower turnover rates indicate oversupply for niche products.

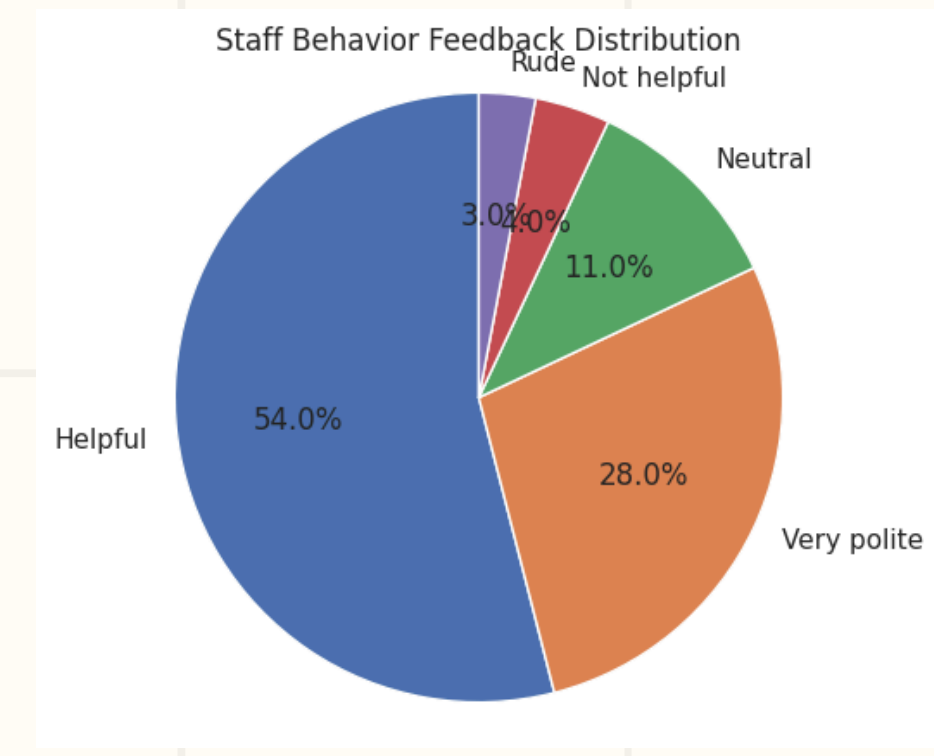
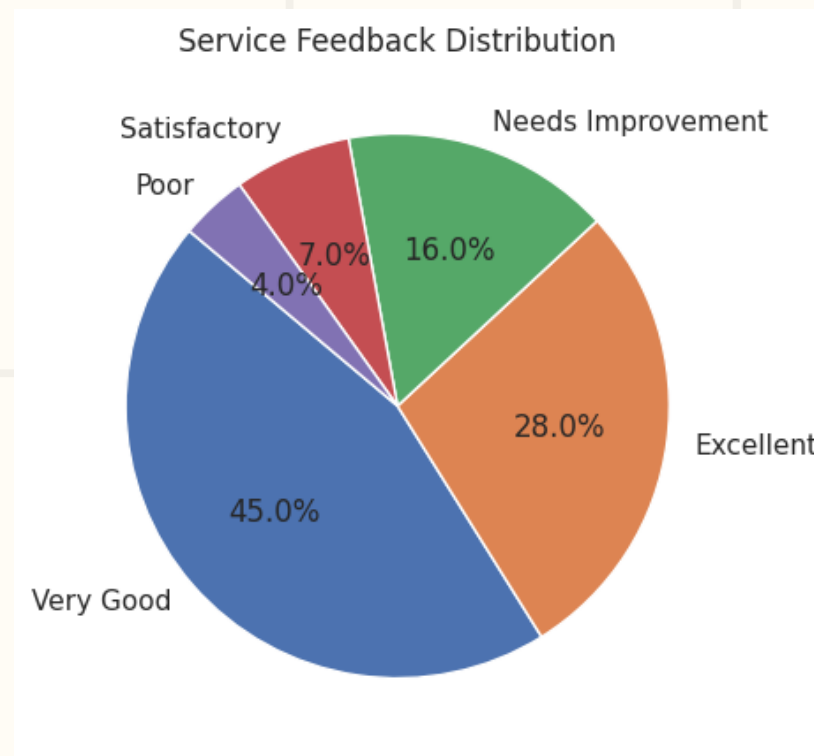
- Inventory Patterns:

High peaks followed by low troughs in inventory levels point to inconsistent stock management, causing inefficiencies in meeting demand.

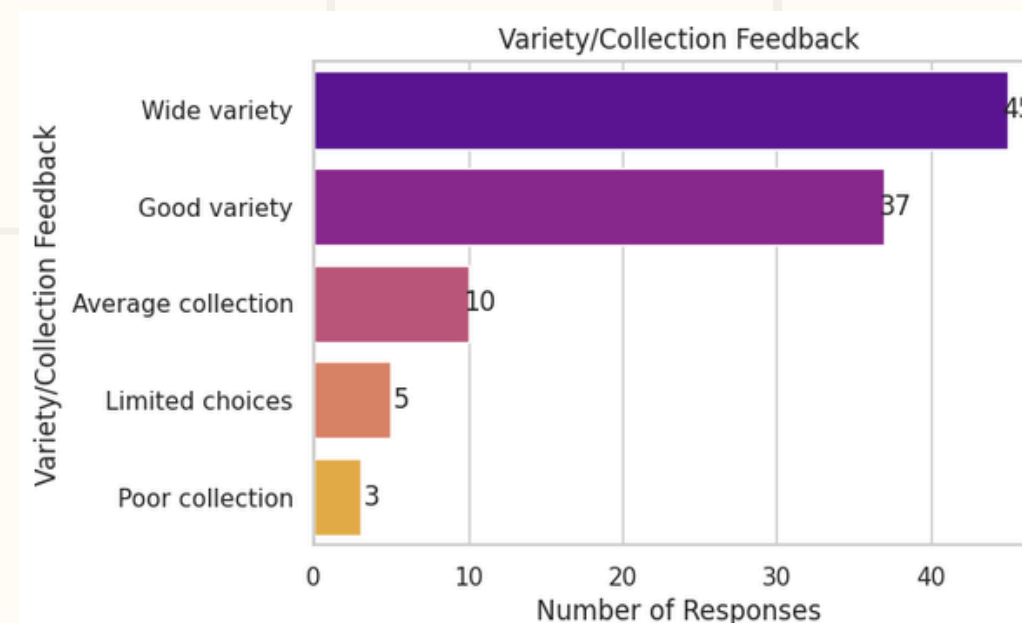


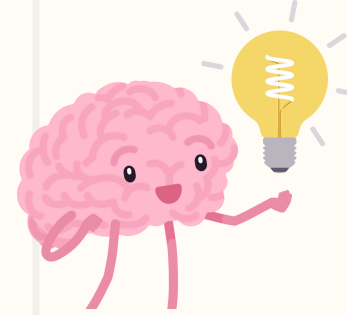
Heatmap of Scaled Turnover Rate and Outwards Quantity





- 73% of customers rated the service as "Very Good" or "Excellent," while 11% indicated areas needing improvement.
- Most deliveries were reported to be 3-5 days late, with few arriving on time or early.
- 82% of customers found staff "Helpful" or "Very Polite," though 7% reported negative experiences.
- Many customers found the store "Messy and unorganized" or "Difficult to find products."
- The majority of customers were satisfied with the variety, with feedback indicating an appealing product range.
- 64% of customers found the pricing "Reasonable" or "Very Affordable," with minor dissatisfaction from price-sensitive groups.
- 67% of customers reported being "Very Satisfied" or "Satisfied," with a small percentage expressing dissatisfaction.





PROPOSED SOLUTIONS

Streamline Client Payments

- Enforce a structured payment policy requiring clients to clear cumulative balances within the following month.
- Introduce penalties:
 - 4% interest for payments delayed beyond the first month.
 - 5% interest for payments delayed beyond the second month.
 - Suspend orders for clients failing to clear dues within three months.
- Why? This ensures timely payments, reduces outstanding balances, and improves cash flow, enabling better financial planning and vendor payment schedules.

Optimize Inventory Management

- Use predictive analytics to forecast demand for high-demand tiles (600x1200 ,24x12) to prevent stockouts.
- Maintain buffer stock for volatile tiles like 800x1600 to avoid missed sales.
- Ensure steady stock for moderate-demand tiles like 100x300 and 1200x1800 to minimize excess inventory.
- Limit inventory for niche tiles like "800x3000" to reduce holding costs.
- Adjust stock levels for seasonal or project-based demand fluctuations.
- Monitor turnover rates to optimize stocking strategies for all tile types.

Enhance Customer Experience

- Address service quality by maintaining high standards
- Improve delivery timelines, as most were reported to be 3-5 days late.
- Reinforce positive staff behavior through training.
- Redesign store layout to resolve issues with organization and navigability.
- Maintain current product variety while exploring additional categories based on feedback.



EXPECTED OUTCOMES

- Improved Cash Flow Management:

Timely payments from clients will reduce outstanding balances and enhance financial stability.

- Optimized Inventory Levels:

Predictive analytics and better stock management will minimize stockouts, reduce excess inventory, and align supply with demand.

- Enhanced Customer Satisfaction:

Improved service quality, timely deliveries, and organized store layouts will address key customer concerns and will foster repeat business and long-term relationships.

- Strengthened Market Position:

Operational efficiency, data-driven decision making and enhanced customer experience will establish the company as a trusted leader in the luxury tiles market.



THANK YOU
DO YOU HAVE ANY
QUESTIONS?

